

Assessment of groundwater quality using GWQI and an AHP-Developed Health-Risk-Oriented Index (HR-GWQI): evidence from the Mitidja Plain (Algeria)

Valutazione della qualità delle acque sotterranee mediante GWQI e un indice orientato al rischio per la salute sviluppato tramite AHP (HR-GWQI): evidenze dalla Pianura della Mitidja (Algeria)

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Abstract

Groundwater quality assessment in the alluvial aquifer system of the Mitidja Plain during the 2019 low-water period was conducted using an integrated approach combining the conventional Groundwater Quality Index (GWQI) with an innovative Human Risk-Groundwater Quality Index (HR-GWQI) developed through Analytic Hierarchy Process (AHP) weighting and GIS-based spatial analysis. Based on eleven physicochemical parameters and data from 23 monitoring wells, the study revealed that natural mineralization driven by carbonate and evaporite dissolution together with seawater intrusion controls the general ionic composition, whereas nitrate contamination of agricultural origin constitutes the main health concern, with 43.5% of samples exceeding the WHO guideline value of 50.0 mg/L. Although the conventional GWQI classified most wells as suitable for drinking purposes, it underestimated nitrate-related health risks due to the uniform weighting of parameters. In contrast, the HR-GWQI, which assigns greater importance to nitrate concentrations through AHP-derived weighting, provided a more realistic and health-protective classification of groundwater quality by reclassifying several contaminated wells into lower quality categories. A sensitivity analysis further confirmed that the HR-GWQI is most robust in the wells with the highest nitrate burden an emergent property of the AHP health-risk-oriented weighting scheme. GIS-based spatial distribution mapping of the monitoring points further revealed that the most significant groundwater quality deterioration is concentrated in the northeastern coastal zone and the western sectors of the plain, areas strongly affected by agricultural pressure and saline intrusion. These findings demonstrate that the HR-GWQI represents a more effective tool for groundwater quality evaluation and health-risk-oriented management in nitrate-contaminated aquifer systems across North Africa and the Mediterranean Basin.

Riassunto

La valutazione della qualità delle acque sotterranee nel sistema acquifero alluvionale della Pianura della Mitidja durante il periodo di magra del 2019 è stata condotta mediante un approccio integrato che combina il convenzionale Groundwater Quality Index (GWQI) con un innovativo Human Risk-Groundwater Quality Index (HR-GWQI), sviluppato attraverso una ponderazione basata sull'Analytic Hierarchy Process (AHP) e un'analisi spaziale mediante GIS. Sulla base di undici parametri fisico-chimici e dei dati provenienti da 23 pozzi di monitoraggio, lo studio ha evidenziato che la mineralizzazione naturale, determinata dalla dissoluzione di carbonati ed evaporiti insieme all'intrusione marina, controlla la composizione ionica generale delle acque sotterranee. La contaminazione da nitrati di origine agricola rappresenta invece la principale criticità per la salute, con il 43,5% dei campioni che supera il valore guida dell'OMS di 50,0 mg/L. Sebbene il GWQI convenzionale abbia classificato la maggior parte dei pozzi come idonei all'uso potabile, esso ha sottostimato i rischi per la salute associati ai nitrati a causa della ponderazione uniforme dei parametri. Al contrario, l'HR-GWQI, che attribuisce maggiore importanza alle concentrazioni di nitrati attraverso una ponderazione derivata dall'AHP, ha fornito una classificazione della qualità delle acque sotterranee più realistica e maggiormente orientata alla tutela della salute, riclassificando diversi pozzi contaminati in categorie di qualità inferiore. Un'analisi di sensitività ha inoltre confermato che l'HR-GWQI è più robusto nei pozzi caratterizzati da concentrazioni di nitrati più elevati, una proprietà emergente dell'approccio di ponderazione AHP orientato al rischio per la salute. La mappatura GIS della distribuzione spaziale dei punti di monitoraggio ha inoltre evidenziato che il deterioramento più significativo della qualità delle acque sotterranee si concentra nella zona costiera nord-orientale e nei settori occidentali della pianura, aree fortemente interessate dalla pressione agricola e dall'intrusione salina. Questi risultati dimostrano che l'HR-GWQI rappresenta uno strumento più efficace per la valutazione della qualità delle acque sotterranee e per una gestione orientata al rischio sanitario nei sistemi acquiferi contaminati da nitrati del Nord Africa e del bacino del Mediterraneo.

Introduction

Groundwater constitutes a critical freshwater resource globally, underpinning agricultural production, industrial development, and the provision of potable water for billions of people (Sanad et al., 2024; Benmarce et al., 2024). The escalating pressures from climate change, rapid population growth, and intensifying anthropogenic activities such as urbanization, industrialization, and intensive agriculture have led to widespread groundwater depletion and quality deterioration, particularly in arid and semi-arid regions like Northern Algeria (Aziez et al., 2025; Ferdjouni et al., 2025; Yahiaoui et al., 2023). In the Mitidja Plain, intensively exploited coastal aquifer in Northern Algeria, these challenges are compounded by overextraction and pollution from both agricultural runoff and urban effluents (Aziez et al., 2025; Khelfi et al., 2025).

Ensuring the safety of groundwater for drinking purposes is paramount for public health. Comprehensive groundwater quality assessment is therefore essential to inform sustainable management strategies (Ferdjouni et al., 2025; Kada et al., 2025). Water Quality Indices (WQIs), especially the Groundwater Quality Index (GWQI), have become widely adopted tools for synthesizing complex hydrochemical data into accessible metrics that facilitate decision-making (Selmane et al., 2023; Jha et al., 2020). GWQI integrates multiple physicochemical parameters to provide an overall assessment of water suitability for human consumption and other uses (Benmarce et al., 2024; Gad et al., 2023).

However, conventional GWQI approaches present notable limitations. Chief among these is the subjective assignment of parameter weights often based on expert judgment rather than objective criteria and a lack of explicit prioritization of health-related risks within the index structure (Jha et al., 2020; Benadela & Bekkousa, 2025). This subjectivity can obscure critical contaminants such as nitrate (NO_3^-), which is frequently overlooked despite its well-documented health impacts. Nitrate contamination, primarily originating from agricultural fertilizers and wastewater infiltration, is recognized as a dominant threat to groundwater quality in Mediterranean and semi-arid aquifers (Kada et al., 2025; Boufekane et al., 2020). Chronic exposure to elevated nitrate levels has been linked to methemoglobinemia (blue baby syndrome) in infants and potential long-term risks including certain cancers (Kada et al., 2025; Houari et al., 2024).

Recent advances have sought to address these shortcomings by integrating multi-criteria decision analysis methods such as the Analytic Hierarchy Process (AHP) (Saaty, 1980) into GWQI frameworks. AHP does not eliminate expert judgment, it decomposes a global judgment into pairwise comparisons, makes the resulting preferences explicit and reproducible, and allows their internal coherence to be tested, thereby providing a structured and auditable basis for weighting (Naz et al., 2024; Benadela & Bekkousa, 2025). Conceptually, the HR-GWQI differs from the conventional index by reframing it from a general suitability metric, where hazardous concentrations can be eclipsed by favorable values of other parameters, into

a health-risk screening tool in which AHP-derived weights reflect recognized health relevance, that guideline exceedance (e.g., nitrate) carries substantially greater weight in the final index value. Furthermore, Geographic Information Systems (GIS) have become indispensable for spatial analysis and mapping of groundwater quality patterns, supporting targeted interventions and resource management at local and regional scales (Abdessamed et al., 2023; Selmane et al., 2023).

Despite these methodological advances, significant research gaps persist. There remains a paucity of studies that explicitly integrate health-risk-oriented prioritization, particularly of nitrate, into GWQI calculations using AHP-based weighting schemes. Comparative analyses between conventional GWQI method by Horton (1965) and Health-Risk-Oriented indices are especially scarce in semi-arid Mediterranean contexts such as the Mitidja Plain (Aziez et al., 2025; Khelfi et al., 2025). Consequently, the sensitivity of GWQI classifications to the weighting scheme itself remains unquantified in these settings, although the compensatory nature of weighted arithmetic aggregation permits an elevated nitrate concentration to be offset by sub-threshold values of the remaining parameters. This paper undertakes a groundwater quality assessment in the Mitidja Plain using both the conventional GWQI (Horton method) and a Health-Risk-Oriented Based Groundwater Quality Index (HR-GWQI) that incorporates AHP-derived weights with explicit prioritization of nitrate and other parameters of recognized health relevance. Eleven key physicochemical indicators including Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} , HCO_3^- , NO_3^- , pH, EC, and TDS were analysed across spatially distributed sampling points. GIS-based spatial mapping of the monitoring network was employed to visualize the distribution of contamination levels at each sampling location.

The contribution of this research lies in its integration of AHP-driven health-risk-oriented prioritization within a GWQI framework, an approach designed to make the weighting explicit and auditable while directly addressing nitrate-related hazards. By systematically comparing traditional GWQI outputs with those from the HR-GWQI model across the Mitidja Plain aquifer system, this study aims to deliver a more health sensitive screening of groundwater quality. The findings are expected to inform evidence-based management strategies for safeguarding public health in vulnerable Mediterranean aquifers.

Materials and Methods

Study Area Description

The Mitidja Plain is a major coastal alluvial basin situated immediately south of Algiers, extending across the cities of Tipaza, Blida, Algiers, and Boumerdès in northern Algeria. The plain has a longitudinal extent of approximately 80-100 km, with a width varying between 8 and 18 km, covering an estimated surface area of $\sim 1,300 \text{ Km}^2$ (Fig.1). It lies at a mean elevation of ~ 100 meters above sea level and is bounded by the Sahel Ridge to the north (~ 260 m), the Blidean Atlas to the south (peaks up to 1,630 m), the Chenoua-Hadjout Massif

to the west, and coastal dunes and low hills to the east toward the Mediterranean Sea.

The climate is typically Mediterranean, characterized by hot, dry summers and warm, wet winters. Regional climatological studies indicate mean annual precipitation between 500 and 700 mm, a mean annual temperature of ~18 °C, and high potential evapotranspiration (Boufekane et al., 2020). These climatic conditions exert strong control on groundwater recharge and seasonal water-table fluctuations.

Geologically, the plain is filled by Neogene to Quaternary deposits overlying older substrata (Fig. 1). The Pliocene is represented by grey-blue marls (Lower Pliocene, Piacenzian) and yellow marls with sandy limestones and molasse facies

(Upper Pliocene, Astian-Villafranchian). Overlying these units, Quaternary alluvium forms thick sequences of sands, gravels, pebbles, and silty-clayey interbeds that infill the basin. These alluvial deposits are highly heterogeneous, with laterally variable thicknesses that can exceed 100 m in some areas, directly influencing the aquifer's hydraulic behavior (Nemer et al., 2023).

From a hydrogeological perspective, the Mitidja Plain hosts two superimposed groundwater systems. Geophysical investigations have confirmed the presence of a confined Astian aquifer (Pliocene) underlying an extensive unconfined Quaternary aquifer (Fig. 2). The latter represents the most productive water resource in the basin, supplying agriculture,

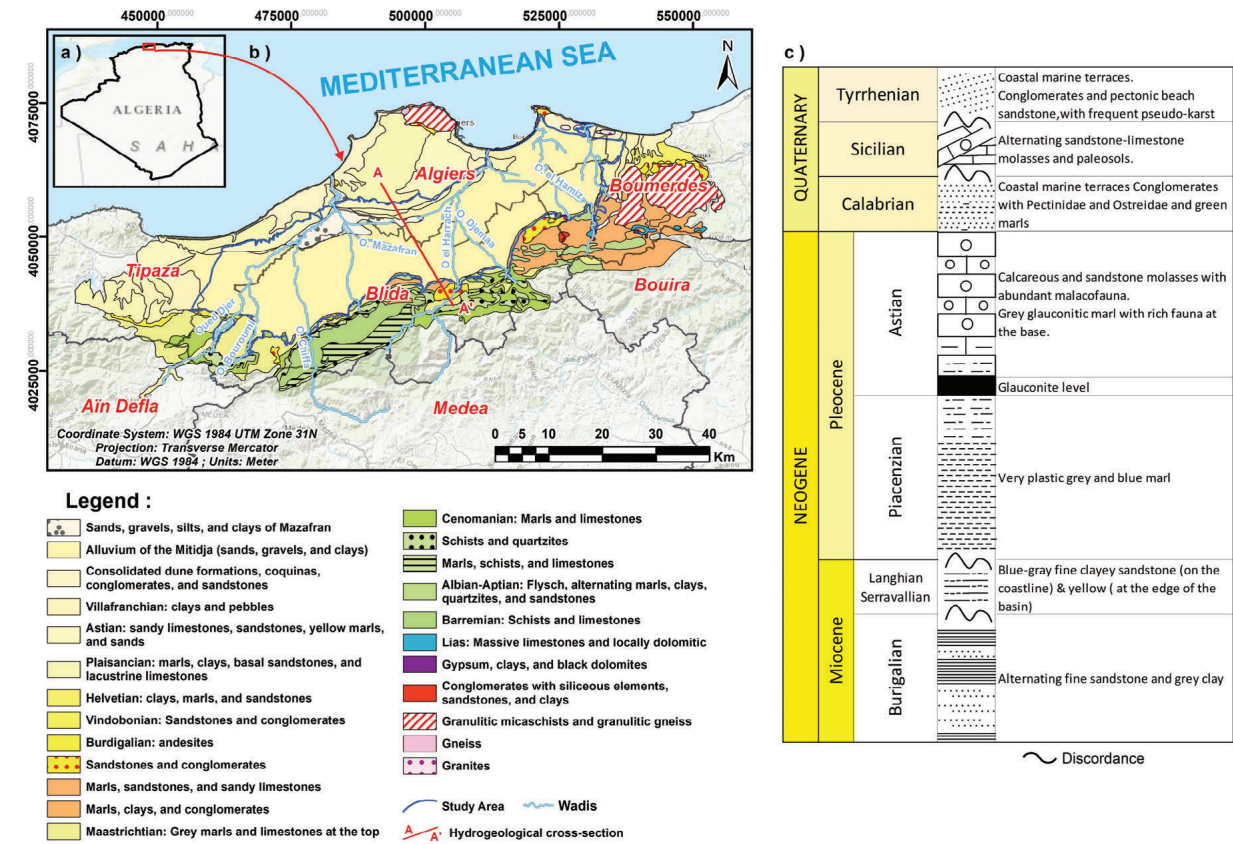


Fig. 1 - Presentation of the Mitidja Plain. a) Location map of the Study Area within Algeria. b) Geological map of the Plain. c) Stratigraphic log of the Neogene and Quaternary deposits (Aymé, 1952; Betrouni, 1983; Djediat, 1996,2011; Glangeaud, 1952; Yassini, 1973).

Fig. 1 - Presentazione della Pianura della Mitidja. a) Carta di localizzazione dell'area di studio in Algeria. b) Carta geologica della pianura. c) Log stratigrafico dei depositi neogenici e quaternari (Aymé, 1952; Betrouni, 1983; Djediat, 1996,2011; Glangeaud, 1952; Yassini, 1973).

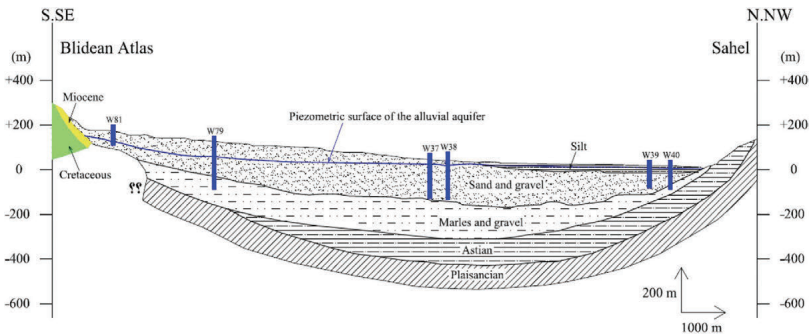


Fig. 2 - Hydrogeological cross-section (AA') (Fig. 1) of the Mitidja Plain (Boufekane et al., 2020).

Fig. 2 - Sezione idrogeologica (AA') (Fig. 1) della Pianura della Mitidja (Boufekane et al., 2020).

industry, and domestic needs. It is composed mainly of fluvial-alluvial sands and gravels, locally resting on marly substrata. Productive horizons within the Quaternary aquifer show variable thickness, from a few tens of meters to ~80-100 m in paleo-channel zones, while the water table remains relatively shallow (4–20 m depth).

Data Collection and Preparation

Groundwater samples were collected by the National Agency of Hydraulic Resources (ANRH) during low-water period (October–November 2019). A total of 23 samples were collected from different locations across the Mitidja aquifer to ensure representative spatial distribution. The sampling protocol involved disinfecting well nozzles, pumping water for several minutes to avoid stagnant effects, and collecting samples in polyethylene containers. All samples were preserved in ice boxes at temperatures below 5 °C and transported directly to the ANRH laboratory for analysis. Field measurements of pH and electrical conductivity were conducted in situ using a multiparameter WTW universal conductivity meter. Total dissolved solids (TDS) were not measured independently via gravimetric analysis; instead, they were estimated directly from the in-situ EC measurements using the standard empirical conversion formula: $\text{TDS (mg/L)} = k \times \text{EC } (\mu\text{S/cm})$, where k is the instrument-specific conversion factor 0.64.

Chemical analyses were performed following standard procedures. Calcium, magnesium, chloride, and bicarbonate were determined by volumetric titrations; sodium and potassium concentrations were measured with a flame photometer, and sulfate by the turbidimetric method. The precision of the analytical work was verified through ionic balance calculations, which generally did not exceed a 5% error. All concentrations were expressed in milligrams per liter (mg/L) unless otherwise indicated. The results were subsequently compared against the World Health Organization (WHO, 2017) and Algerian (JORA, 2011) water quality standards.

Calculation of Groundwater Quality Index (GWQI)

Groundwater quality was assessed using two complementary indices: a conventional Weighted Arithmetic Index (Horton, 1965) and a health-risk-oriented variant with AHP-derived weights (HR-GWQI), enabling both standard physicochemical and toxicologically informed evaluation of drinking water suitability.

a) Conventional GWQI (Horton Method)

The GWQI was computed via the Weighted Arithmetic Index Method (Horton, 1965; Brown et al., 1970) in four steps.

Step 1 – Relative Weights: Each parameter was assigned an expert weight (w_i , 1–5) reflecting its drinking water significance. The relative weight was normalized as:

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (1)$$

Weights and WHO guideline values are listed in (Table 1)

Step 2 – Quality Rating (q_i): For parameters exhibiting a monotonic concentration–quality relationship (e.g., major ions, NO_3^- , TDS, EC), the quality rating was calculated as the percentage ratio of measured concentration (C_i) to the WHO guideline value (S_i) (WHO, 2017):

$$q_i = \frac{C_i}{S_i} \cdot 100 \quad (2)$$

For pH whose risk is non-monotonic (i.e., both acidic and alkaline shifts away from neutrality degrade water quality), the standard linear formulation is not applicable. The modified formulation of Akter et al. (2016) is applied: $q_i = |\text{pH}_{\text{meas}} - 7.0| / (8.5 - 7.0) \times 100$, which captures the deviation from the optimal neutral value (7.0) toward either the acidic or the alkaline side.

Step 3 – Sub-Index (SI_i):

$$SI_i = W_i \cdot q_i \quad (3)$$

Step 4 – Final GWQI Aggregation:

$$\text{GWQI} = \sum_{i=1}^n SI_i \quad (4)$$

Tab. 1 - WHO Guideline Values, Expert Weights, and Relative Weights for the Conventional GWQI (Horton, 1965; WHO, 2017).

Tab. 1 - Valori guida dell'OMS, pesi degli esperti e pesi relativi per il GWQI convenzionale (Horton, 1965; OMS, 2017).

Parameter	WHO Standard (S_i , mg/L)	Weight (w_i)	Relative Weight (W_i)
Ca^{2+} (mg/L)	75	3	0.0714
Mg^{2+} (mg/L)	50	3	0.0714
Na^+ (mg/L)	200	5	0.1190
K^+ (mg/L)	12	2	0.0476
Cl^- (mg/L)	250	5	0.1190
SO_4^{2-} (mg/L)	250	3	0.0714
HCO_3^- (mg/L)	500	3	0.0714
NO_3^- (mg/L)	50	5	0.1190
pH	6.5–8.5 (ideal: 7.0)	5	0.1190
EC ($\mu\text{S/cm}$)	2.000	4	0.0952
TDS (mg/L)	1.000	4	0.0952
Σ	—	42	1.0000

b) Health-Risk-Oriented GWQI (HR-GWQI) via AHP

The Horton method does not account for differential toxicological significance. In the Mitidja Plain, NO_3^- driven by intensive agricultural fertilization is the dominant contaminant, associated with infant methemoglobinemia, endocrine disruption, and probable carcinogenicity (WHO, 2017). To address this, parameter weights were replaced by priorities derived from Saaty's Analytic Hierarchy Process (AHP) (Saaty, 1980, 1987).

b.1. Pairwise Comparison Matrix

An 11×11 matrix was constructed for the parameters Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} , HCO_3^- , NO_3^- , pH, EC, and TDS using Saaty's 1–9 scale (Table 2). NO_3^- received the highest priority; Na^+ and Cl^- were ranked secondary due to cardiovascular and salinity concerns; SO_4^{2-} , TDS, and EC received intermediate weights for their mineralization and gastrointestinal effects; pH was moderately weighted; Ca^{2+} , Mg^{2+} , HCO_3^- , and K^+ received the lowest weights as essential minerals with limited toxicity at natural concentrations (Minh et al., 2019; Badihi et al., 2023). The matrix satisfies diagonal unity ($a_{ii} = 1$) and reciprocity ($a_{ij} = 1/a_{ji}$) (Fig. 3).

Tab. 2 - Saaty's Fundamental Scale of Relative Importance (Saaty, 1980).

Tab. 2 - Scala fondamentale di importanza relativa di Saaty (Saaty, 1980).

Score	Verbal Judgement	Explanation
1	Equal importance	Two criteria contribute equally to the objective
2	Weak or slight	
3	Moderate importance	Experience and judgement slightly favour one over another
4	Moderate plus	
5	Strong importance	Experience and judgement strongly favour one over another
6	Strong plus	
7	Very strong importance	A criterion is favoured very strongly; its dominance is demonstrated in practice
8	Very, very strong	
9	Extreme importance	The evidence favouring one over another is of the highest possible order

b.2. Priority Vector (Eigenvector Method)

Following the construction of the pairwise comparison matrix ($n \cdot n$), the relative weights, also referred to as the priority vector or normalised weight vector $\mathbf{W}$, were derived using Saaty's principal right eigenvector method (Saaty, 1977, 2003). This approach assumes that, for a perfectly consistent reciprocal matrix, the priority vector satisfying $\mathbf{A} \cdot \mathbf{w} = \lambda_{\max} \cdot \mathbf{w}$ is unique (to a positive multiplicative constant), where $\lambda_{\max}$ is the maximum eigenvalue of $\mathbf{A}$ (Saaty, 2003). The calculation proceeds through the following standardised steps:

Step EV-1 – Normalisation of Each Column. Each element a_{ij} of the pairwise comparison matrix is divided by the sum of its column to obtain the normalised matrix $\bar{\mathbf{A}} = [\bar{a}_{ij}]$:

$$\bar{a}_{ij} = \frac{a_{ij}}{\sum_{m=1}^n a_{mj}} \quad (5)$$

Step EV-2 – Row Averaging (Priority Vector). The relative weight W_i for each parameter i is computed as the arithmetic mean of the i -th row of the normalised matrix $\bar{\mathbf{A}}$:

$$W_i = \frac{1}{n} \cdot \sum_{j=1}^n \bar{a}_{ij} \quad (6)$$

The resulting vector $\mathbf{W} = (W_1, W_2, \dots, W_n)^T$ constitutes the priority vector (approximate principal right eigenvector) of matrix $\mathbf{A}$. By construction, $\sum_i W_i = 1$, ensuring that the weights are normalised and collectively represent 100% of the total criterion importance (Escobar et al., 2004) (Table 3). This approximation, known as the means of normalized values (MNV) or additive normalization (AN) method, is widely applied in practice due to its computational simplicity and acceptable accuracy when the matrix is sufficiently consistent (Saaty, 2003).



Fig. 3 - AHP Pairwise Comparison Matrix for Health-Risk-Oriented Parameter Weighting (Bisen & Singh, 2026; Saaty, 1987).

Fig. 3 - Matrice di confronto a coppie AHP per la ponderazione dei parametri orientata al rischio per la salute (Bisen & Singh, 2026; Saaty, 1987).

Tab. 3 - AHP-Derived Priority Weights (Eigenvector W_i) for the HR-GWQI Parameters.

Tab. 3 - Pesi di priorità derivati dall'AHP (autovettore W_i) per i parametri dell'HR-GWQI.

Parameter	Indicator	Priority Weight (W_i)
NO_3^- (mg/L)	Nitrate	32.00%
Na^+ (mg/L)	Sodium	13.81%
Cl^- (mg/L)	Chloride	9.94%
SO_4^{2-} (mg/L)	Sulfate	9.33%
TDS (mg/L)	Total Dissolved Solids	8.84%
EC ($\mu\text{S}/\text{cm}$)	Electrical Conductivity	8.03%
pH	Potential of hydrogen	6.23%
Ca^{2+} (mg/L)	Calcium	3.40%
Mg^{2+} (mg/L)	Magnesium	3.40%
HCO_3^- (mg/L)	Bicarbonate	3.28%
K^+ (mg/L)	Potassium	1.73%
Σ	—	100.00%

b.3. Determination of the Maximum Eigenvalue ($\lambda_{\max}$)

The maximum eigenvalue $\lambda_{\max}$ is computed to evaluate the internal consistency of the pairwise judgements. For a perfectly consistent $n \cdot n$ matrix, $\lambda_{\max} = n$ exactly; deviations from n indicate the presence of judgment inconsistencies. $\lambda_{\max}$ is estimated from the priority vector W and the matrix A as follows:

$$\lambda_{\max} = \frac{1}{n} \cdot \sum_i \frac{(A \cdot W)_i}{w_i} \quad (7)$$

where $(A \cdot W)_i$ denotes the i -th element of the matrix-vector product $A \cdot W$ (i.e., the weighted sum of the i -th row of A). For the HR-GWQI pairwise matrix in this study, $\lambda_{\max} = 11.643$.

b.4. Assessment of Matrix Consistency

The reliability of AHP-derived weights depends critically on the internal consistency of the pairwise judgements. Saaty (1990) developed two complementary metrics for this purpose.

- **Consistency Index (CI):** The CI measures the deviation of $\lambda_{\max}$ from n , where larger deviations indicate greater inconsistency:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (8)$$

For $n=11$ and $\lambda_{\max}=11.643$:

$$CI = (11.643 - 11) / (11 - 1) = 0.643 / 10 = 0.0643.$$

- **Consistency Ratio (CR):** The CR normalizes the CI against the Random Index (RI), which is the expected CI for a randomly generated reciprocal matrix of the same dimension (Saaty, 1987). The formula is:

$$CR = \frac{CI}{RI} \quad (9)$$

The RI values for matrices of different dimensions are given in (Table 4). For $n=11$, $RI=1.51$.

Tab. 4 - Saaty's Random Consistency Index (RI) Values for different matrix dimensions (Saaty, 1987).

Tab. 4 - Valori dell'indice di consistenza casuale (RI) di (Saaty, 1987) per diverse dimensioni delle matrici.

n	1	2	3	4	5	6	7	8	9	10	11
RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51

Substituting the calculated values: $CR = 0.0643 / 1.51 = 0.0426$ (4.26%). Since $CR < 0.10$ (the widely accepted threshold for adequate consistency; Saaty, 1980), the pairwise judgements are deemed logically coherent, and the derived weights are considered reliable for application in the health-risk-oriented groundwater quality index.

b.5. Computation of the Health-Risk-Oriented GWQI (HR-GWQI)

The HR-GWQI was computed using the same weighted arithmetic aggregation structure as the conventional GWQI (Eqs 3 and 4) but substituting the Horton expert-assigned relative weights (W_i) with the AHP-derived priority weights (W_i^{AHP}). The two-step procedure is as follows:

Step HR-1 – Health-Risk Sub-Index (HRSI_i):

The sub-index for each parameter i is computed using the quality rating q_i from Equation (2) and the AHP-derived weight:

$$HRSI_i = W_i^{AHP} \cdot q_i \quad (10)$$

Step HR-2 – Final HR-GWQI Aggregation:

The overall HR-GWQI is obtained by summing all parameter health-risk sub-indices:

$$HR-GWQI = \sum_i HRSI_i = \sum_i (W_i^{AHP} \cdot q_i) \quad (11)$$

To assess the robustness of the HR-GWQI to the mathematical dependency between EC and TDS, a sensitivity analysis was performed by removing TDS from the weighting scheme and redistributing its weight ($W_{TDS} = 8.84\%$) proportionally among the remaining ten parameters using:

$$W'_i = W_i / (1 - W_{TDS}) \quad (12)$$

The revised HR-GWQI was computed as:

$$GWQI_{HR}^* = \frac{[GWQI_{HR} - W_{TDS} \cdot q_{TDS}]}{(1 - W_{TDS})} \quad (13)$$

This formulation yields a groundwater quality index that explicitly emphasizes parameters with greater documented or potential impacts on human health, thereby providing a toxicologically more representative assessment of drinking water suitability than the conventional GWQI.

Both GWQI and HR-GWQI values were subsequently classified according to the five-category scheme proposed by Sadat-Noori et al. (2014), as presented in (Table 5).

Spatial Distribution Mapping

Spatial distribution maps of groundwater quality parameters, GWQI, and HR-GWQI were produced using ArcMap 10.8 by mapping each monitoring well as a classified point symbol proportional to its measured value or index

Tab. 5 - Classification of Groundwater Quality Based on GWQI and HR-GWQI Values (Sadat-Noori et al., 2014).

Tab. 5 - Classificazione della qualità delle acque sotterranee basata sui valori di GWQI e HR-GWQI (Sadat-Noori et al., 2014).

GWQI / HR-GWQI Range	Water Quality Category	Interpretation
< 50	Excellent	Suitable for all uses; no treatment required
50–75	Good – Level 1	Good water quality; generally suitable for drinking
75–100	Good – Level 2	
100–125	Poor – Level 1	Slightly degraded quality; treatment may be required before consumption
125–150	Poor – Level 2	Poor quality; treatment required before consumption
150–175	Poor – Level 3	Poor quality; substantial treatment required before consumption
175–200	Poor – Level 4	Very poor end of the poor-quality range; extensive treatment required
200–225	Very Poor – Level 1	Very poor quality; unsuitable for domestic use without extensive treatment
225–250	Very Poor – Level 2	Very poor quality; extensive treatment required
250–275	Very Poor – Level 3	Severely degraded quality; generally unsuitable for domestic use
275–300	Very Poor – Level 4	Extremely degraded quality; unsuitable for domestic use
> 300	Unsuitable for Drinking	Highly contaminated; unfit for human consumption

score. This point-based visualization approach was used to identify spatial patterns, contamination hotspots, and quality classification across the Mitidja Plain without assuming continuous gradients between sampling locations. All generated maps were projected using the UTM Zone 31N coordinate system.

Results

Descriptive statistics of hydrochemical parameters

Statistical analysis of eleven physicochemical parameters across 23 sampling points (Table 6) reveals pronounced

spatial heterogeneity in groundwater quality, reflecting the combined influence of lithological controls, marine intrusion, and anthropogenic pressures.

Major cations exhibit elevated concentrations relative to WHO guidelines: Ca^{2+} (mean: 159.21 mg/L; 95.7% exceedance) and Mg^{2+} (39.10 mg/L; 21.7% exceedance) primarily derive from the dissolution of Miocene–Pliocene carbonate and dolomitic formations. Na^+ concentrations (mean: 95.65 mg/L) approach regulatory limits in coastal sectors, consistent with seawater intrusion in over-exploited aquifers, while K^+ remains negligible (mean: 1.34 mg/L) and within acceptable thresholds.

Tab. 6 - Descriptive Statistics of hydrochemical parameters.

Tab. 6 - Statistiche descrittive dei parametri idrochimici.

Parameter	Mean	Min	Max	Std Dev	WHO Limit	Exceedance
Ca^{2+} (mg/L)	159,21	19,6	278,43	54,69	75	95.7%
Mg^{2+} (mg/L)	39,1	11,76	82,35	18,25	50	21.7%
Na^+ (mg/L)	95,65	25	209	59,81	200	8.7%
K^+ (mg/L)	1,34	0	12	2,58	12	0.0%
Cl^- (mg/L)	152,61	38	360	97,88	250	17.4%
SO_4^{2-} (mg/L)	180,65	0	370	96,73	250	17.4%
HCO_3^- (mg/L)	405,78	137,25	732	134,82	500	26.1%
NO_3^- (mg/L)	50,57	0	125	30,45	50	43.5%
pH	7,79	7,5	8,3	0,18	6.5–8.5	0.0%
EC ($\mu\text{S}/\text{cm}$)	1459,43	445	2490	555,69	2000	17.4%
TDS (mg/L)	1103,75	318,58	1888,77	426,83	1000	60.9%

Among anions, Cl^- (mean: 152.61 mg/L) and SO_4^{2-} (180.65 mg/L) show moderate exceedance rates (17.4%), with elevated Cl^- in coastal zones reinforcing evidence of saline encroachment. HCO_3^- concentrations (mean: 405.78 mg/L; 26.1% exceedance) reflect active water–rock interaction in carbonate-rich media, contributing to the system's alkaline buffering capacity. Nitrate levels display marked variability (0–125 mg/L; mean: 50.57 mg/L), with 43.5% of samples exceeding the 50.0 mg/L drinking water standard, indicating significant contamination from agricultural runoff and wastewater discharge.

Physical parameters confirm moderate to high mineralization: pH values (7.50–8.30) remain within acceptable limits, denoting stable carbonate-buffered conditions; electrical conductivity (mean: 1459.43 $\mu\text{S}/\text{cm}$) and TDS (mean: 1103.75 mg/L; 60.9% exceedance) reflect the same underlying ionic mineralization and spatially aligned with Na^+ – Cl^- enrichment, underscoring salinization as the principal driver of groundwater degradation.

Quality Classification: Comparative Performance of GWQI-Horton and HR-GWQI- AHP

Comparative Assessment and Health-Risk Sensitivity of GWQI and HR-GWQI

The comparative analysis between the conventional Groundwater Quality Index (GWQI) and the AHP-based Health-Risk-Oriented Groundwater Quality Index (HR-GWQI) demonstrates significant differences in groundwater quality classification and sensitivity to health-related parameters. Overall, the conventional GWQI indicates a predominance of “good water quality,” with several samples classified as “excellent” and only a limited number categorized as “poor.” In contrast, the HR-GWQI produces a more conservative assessment, increasing the number of samples classified as “poor water quality.” This shift reflects the influence of the AHP weighting scheme, which prioritizes parameters associated with human health risks, particularly nitrate (NO_3^-) and sodium (Na^+). As a result, several wells initially classified as “good” under the conventional GWQI were downgraded under the HR-GWQI framework, including samples with GWQI values of 98.89, 91.13, and 85.13 that increased to HR-GWQI values of 118.00, 102.82, and 101.08, respectively, thereby shifting from the “Good” to the “Poor” category. These reclassifications demonstrate that the conventional GWQI tends to underestimate potential health risks because it distributes weights more uniformly among parameters, whereas the HR-GWQI selectively amplifies the contribution of toxicologically significant contaminants, resulting in classifications more consistent with epidemiological risk thresholds and public health protection objectives. The calculated GWQI values ranged from 29.65 to 131.58, with a mean value of 80.99 ± 27.45 , while HR-GWQI values ranged from 20.94 to 144.73, with a mean of 83.42 ± 29.18 . Application of both indexing methods to the 23 monitoring wells revealed systematic reclassification patterns, confirming that the HR-GWQI framework provides a more

health-oriented and precautionary evaluation of groundwater quality.

Groundwater Quality Index Assessment

The comparative evaluation of the conventional Horton-based GWQI and the AHP-derived HR-GWQI across the 23 monitoring wells demonstrates how explicitly prioritizing human health risks particularly nitrate toxicity and cumulative ionic load fundamentally alter groundwater quality classifications within the Mitidja Plain (Figs 4, 5, and 6).

The calculated GWQI values ranged from 29.65 to 131.58, with a mean of 80.99 ± 27.45 . Based on the modified classification system, the groundwater quality distribution was as follows: Excellent (<50): 3 samples (13.0%); Good level 1 (50–75): 7 samples (30.4%); Good level 2 (75–100): 8 samples (34.8%); Poor level 1 (100–125): 4 samples (17.4%); Poor level 2 (125–150): 1 sample (4.3%). Aggregating the Good categories, 65.2% of samples fell within the 50–100 range, yielding a total of 78.3% of wells classified as suitable for drinking purposes without treatment.

In contrast, the HR-GWQI values ranged from 20.94 to 144.73, with a mean of 83.42 ± 29.18 , indicating slightly greater dispersion attributable to the differential weighting of high-risk parameters. Under the same classification thresholds, the HR-GWQI distribution was: Excellent (<50): 3 samples (13.0%); Good (50–100): 13 samples (56.5%); Poor (100–150): 7 samples (30.4%). Consequently, the proportion of samples deemed suitable for consumption decreased from 78.3% under GWQI to 69.6% under HR-GWQI, reflecting the index's heightened sensitivity to toxicologically relevant contaminants.

These results indicate that 69.6% of groundwater samples are suitable for drinking purposes under the health-risk-weighted index, compared to 78.3% under the standard GWQI. The HR-GWQI thus classifies an additional 8.7% of samples as requiring treatment, reflecting its heightened sensitivity to parameters with direct toxicological relevance.

Reclassification analysis identified three wells (13.0%) that shifted from “Good” under GWQI to “Poor” under HR-GWQI: W021-417 ($\Delta = +19.12$), W042-439 ($\Delta = +11.69$), and W042-1003 ($\Delta = +15.95$). All three exhibited elevated nitrate concentrations (87–112 mg/L) coincident with high electrical conductivity (>1,800 $\mu\text{S}/\text{cm}$) and total dissolved solids (>1,100 mg/L). The HR-GWQI's amplified weighting of nitrate and salinity parameters elevated their composite scores beyond the “Good” threshold (100), triggering reclassification despite moderate performance on less hazardous ions.

Conversely, one well (4.3%) shifted from “Poor” under GWQI to “Good” under HR-GWQI: W021-893 (GWQI = 112.53; HR-GWQI = 90.53; $\Delta = -22.00$). This well displayed elevated calcium and bicarbonate concentrations but relatively low nitrate (<30 mg/L); the HR-GWQI's down weighting of these benign major ions reduced the composite score.

The mean absolute difference between indices was 9.9 ± 11.2 points. The largest positive discrepancies (>+15 points) occurred in wells where nitrate exceeded 80 mg/L or ionic

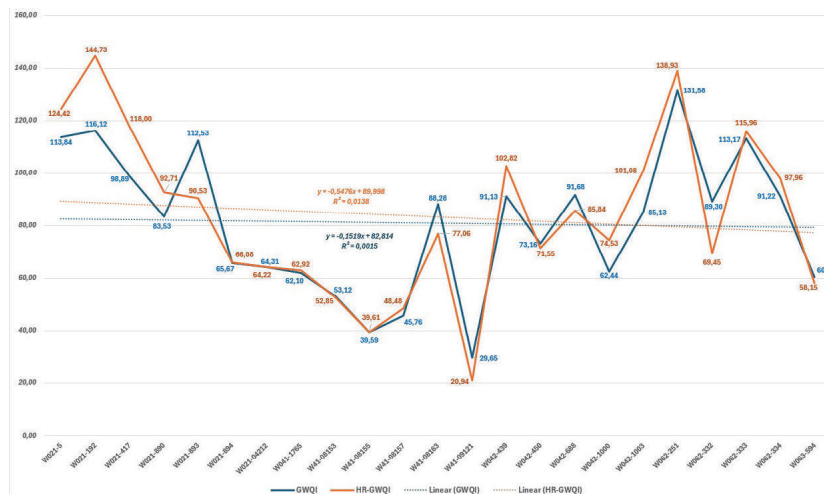


Fig. 4 - Comparative variation of conventional GWQI and HR-GWQI-AHP values.

Fig. 4 - Variazione comparativa dei valori del GWQI convenzionale e dell'HR-GWQI-AHP.

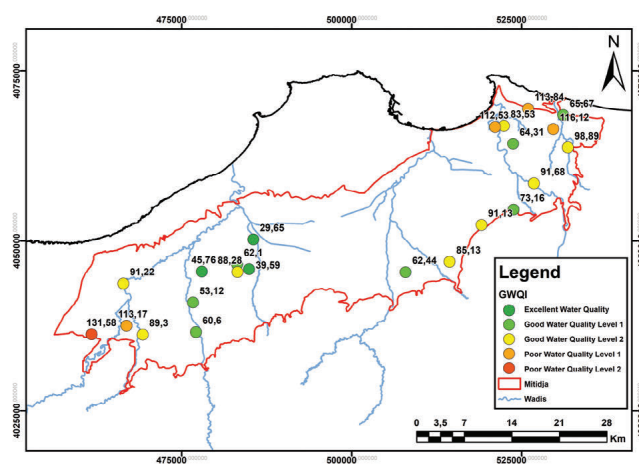


Fig. 5 - The spatial distribution of the GWQI by Horton method in the Mitidja Plain during the low-water period 2019.

Fig. 5 - Distribuzione spaziale del GWQI calcolato con il metodo di Horton nella Pianura della Mitidja durante il periodo di magra del 2019.

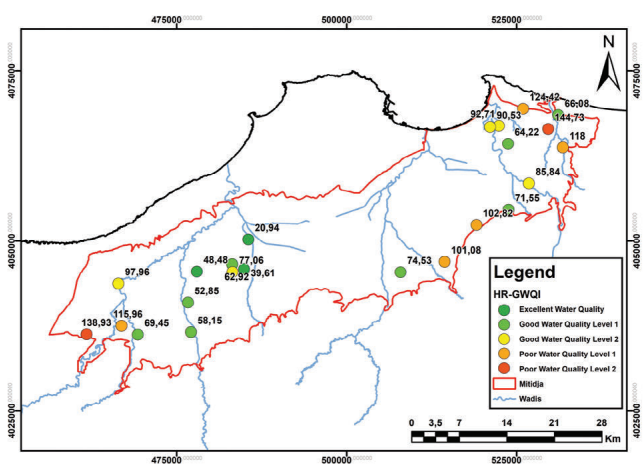


Fig. 6 - The spatial distribution of the Modified HR-GWQI by AHP method in the Mitidja Plain during the low-water period 2019.

Fig. 6 - Distribuzione spaziale dell'HR-GWQI modificato mediante metodo AHP nella Pianura della Mitidja durante il periodo di magra del 2019.

mineralization surpassed guideline thresholds ($EC > 2,000$ $\mu S/cm$ and $TDS > 1,000$ mg/L), whereas the largest negative discrepancies (-11 to -22 points) corresponded to wells dominated by non-toxic major ions. These patterns confirm that the HR-GWQI does not uniformly inflate or deflate scores but selectively amplifies the influence of health-critical parameters. The HR-GWQI does not distort extreme conditions but rather refines intermediate classifications where risk interpretation is more uncertain.

Correlation analysis and hydrochemical processes

Pearson correlation analysis (Fig. 7) elucidates the dominant geochemical controls on groundwater composition in the Mitidja Plain. As expected from the empirical conversion used during analysis, EC and TDS exhibit a deterministic linear relationship ($r = 1.00$). Because this correlation is a mathematical artifact of the calculation method rather than an independent hydrochemical relationship, it is not interpreted as a distinct geochemical process. Instead, EC and TDS are jointly considered as indicators of bulk mineralization.

Calcium exhibits robust correlations with both EC and TDS ($r = 0.88$), underscoring its dominant contribution to water hardness and overall salinity. The pronounced Na^+-Cl^- relationship ($r = 0.90$), reinforced by strong Na^+ linkages to EC and TDS ($r = 0.86$), indicates a common saline source, consistent with marine intrusion or evaporite dissolution.

Moderate correlations further delineate specific geochemical pathways: Ca^{2+} associates significantly with SO_4^{2-} ($r = 0.82$) and HCO_3^- ($r = 0.71$), reflecting the concurrent dissolution of evaporitic (gypsum) and carbonate minerals. Conversely, nitrate displays negligible correlations with major ions, confirming an independent anthropogenic origin driven by agricultural runoff and wastewater discharge. pH exhibits consistent negative correlations with dissolved species, reflecting the expected inverse relationship between acidity and mineralization in carbonate-buffered systems.

Collectively, these correlation patterns identify three principal hydrochemical processes governing aquifer chemistry: (1) water-rock interaction, evidenced by $Ca^{2+}-HCO_3^-$ and $Ca^{2+}-SO_4^{2-}$ associations; (2) salinization,

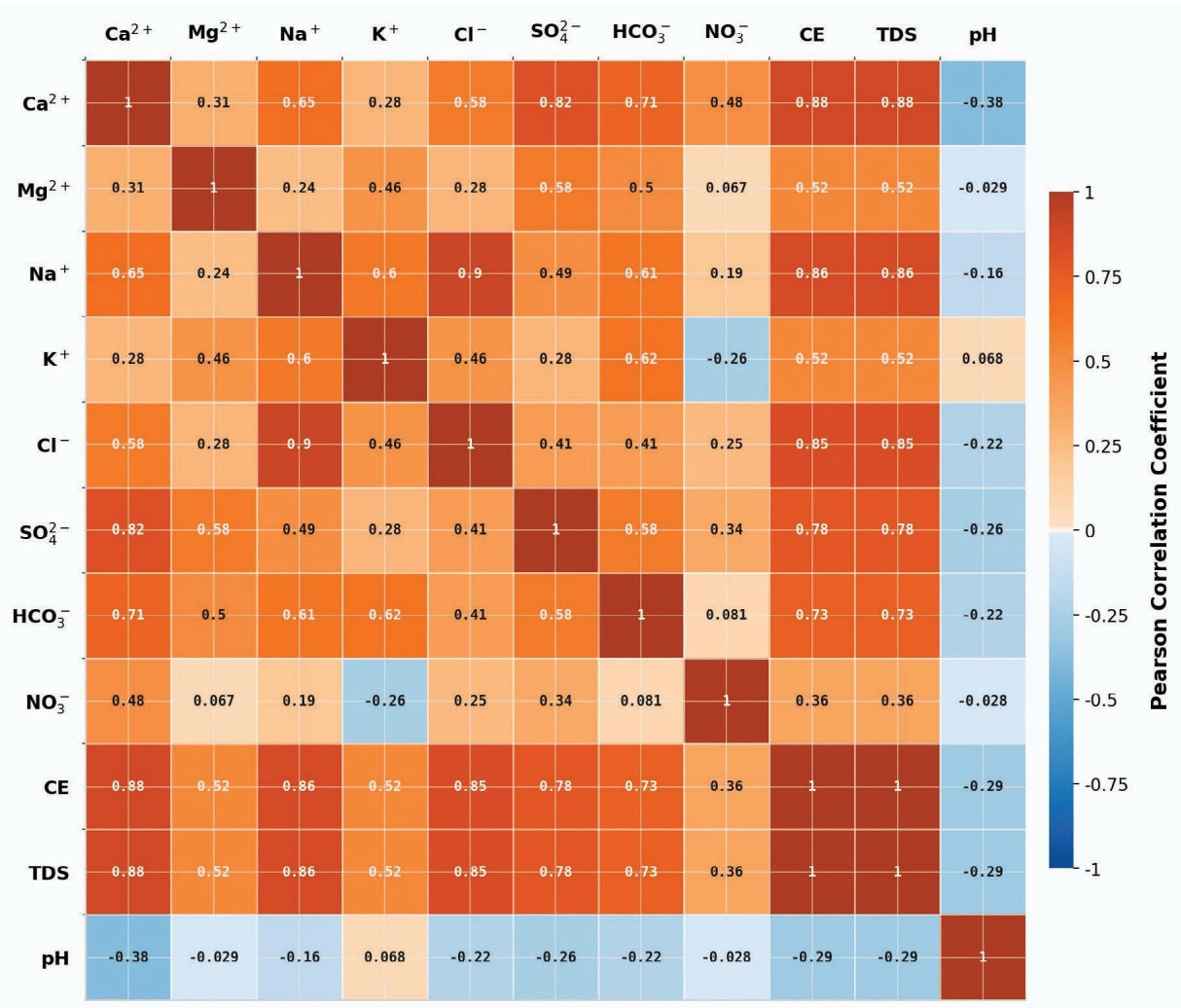


Fig. 7 - Correlation matrix between groundwater quality parameters of Mitidja Plain.

Fig. 7 - Matrice di correlazione tra i parametri della qualità delle acque sotterranee nella Painura Mitidja.

driven by Na⁺–Cl⁻ enrichment and their strong linkage to EC/TDS; and (3) anthropogenic contamination, characterized by the decoupled behaviour of NO₃⁻ from natural ion suites.

Discussion and Conclusion

Discussion

The hydrochemical characterization of the Mitidja Plain aquifer during the low-water period of 2019 reveals a groundwater system affected by both natural geochemical processes and anthropogenic contamination, highlighting the limitations of conventional weighted arithmetic indices in evaluating groundwater quality where contaminants of unequal toxicological significance coexist. The dominant hydrochemical signature across the 23 monitoring wells is characterized by moderate to high mineralization. EC and TDS co-vary by mathematical construction ($r = 1.00$) and are not interpreted as an independent hydrochemical relationship. The significant correlation between Ca²⁺ and EC ($r = 0.88$) indicates that groundwater salinization is mainly controlled by carbonate and evaporite dissolution as well as

seawater intrusion in coastal areas. Similarly, the strong Na⁺–Cl⁻ correlation ($r = 0.90$) confirms saline intrusion processes linked to chronic groundwater overexploitation, as previously reported by Khelfi et al. (2025) and Ferdjouni et al. (2025). Additional correlations between Ca²⁺–SO₄²⁻ ($r = 0.82$) and Ca²⁺–HCO₃⁻ ($r = 0.71$) indicate the simultaneous influence of gypsum and calcite dissolution within a carbonate-buffered hydrochemical environment.

In contrast to the major ions, nitrate exhibits a distinct hydrochemical behaviour. Its absence of correlation with the natural ionic suite demonstrates that nitrate contamination is not related to geogenic mineralization processes but originates primarily from diffuse agricultural activities associated with intensive horticulture and, secondarily, wastewater infiltration in peri-urban sectors (Boufekane et al., 2020; Aziez et al., 2025). Nitrate concentrations exceeded the WHO guideline value of 50 mg/L in 43.5% of samples and reached a maximum of 125 mg/L, making nitrate the most critical contaminant from a public health perspective.

These hydrochemical conditions expose the structural

limitations of the conventional Horton (1965) Groundwater Quality Index (GWQI). The Horton approach distributes weights among parameters using expert judgment without a formal mathematical validation criterion, assigning equal maximum weights to Na^+ , Cl^- , NO_3^- , and pH despite their markedly different toxicological implications. This assumption is scientifically problematic because nitrate concentrations above the WHO threshold can induce methemoglobinemia and long-term carcinogenic risks, whereas sodium and chloride generally exert lower physiological impacts at comparable concentrations, and the pH values observed in the study area (7.50–8.30) pose no direct health hazard (WHO, 2017; Kada et al., 2025). Moreover, the arithmetic aggregation used in GWQI dilutes the contribution of highly toxic contaminants within the overall index. Consequently, wells W021-417, W042-439, and W042-1003 were classified as “Good” under the conventional GWQI despite nitrate concentrations ranging between 87 and 112 mg/L, clearly exceeding safe drinking-water limits. Conversely, well W021-893 was classified as “Poor” mainly due to elevated Ca^{2+} and HCO_3^- concentrations, although these ions are generally considered physiologically beneficial and its nitrate concentration remained below 30 mg/L.

To address these shortcomings, the Human Risk-Groundwater Quality Index (HR-GWQI) was developed using an AHP-based weighting framework derived from the methodology of Saaty (1977, 1980, 1987). The AHP approach applies systematic pairwise comparisons using Saaty’s 1–9 scale and evaluates the logical coherence of judgments through the Consistency Ratio (CR). The CR obtained in this study (0.0426) is well below the acceptable threshold of 0.10, confirming the mathematical reliability of the weighting scheme (Escobar et al., 2004; Rajkumar et al., 2022). Although TDS was estimated from EC, both parameters are retained as separate criteria following the convention of Horton (1965) and consistent with established GWQI practice (Naz et al., 2024; Gad et al., 2023; Benadela & Bekkousa, 2025). Following Saaty (1980), the independence condition in AHP refers to preference independence, not measurement independence; EC and TDS carry independently defined WHO guideline values (2,000 $\mu\text{S}/\text{cm}$ and 1,000 mg/L respectively) and were assigned distinct expert judgements in the pairwise matrix. The sensitivity analysis further reveals a structurally significant property of the HR-GWQI that reinforces the validity of the AHP weighting scheme. When TDS is removed from the index and its weight redistributed proportionally among the remaining parameters, the magnitude of the resulting HR-GWQI variation is inversely related to the nitrate concentration of each well. In wells with elevated nitrate burdens such as W021-192 ($\text{NO}_3^- = 125 \text{ mg/L}$, $\Delta = +0.43$) and W021-417 ($\text{NO}_3^- = 92 \text{ mg/L}$, $\Delta = -0.71$) the contribution of TDS to the composite index is marginal relative to that of NO_3^- , and the HR-GWQI is virtually insensitive to the treatment of TDS. Conversely, in wells with low nitrate concentrations such as W021-893 ($\text{NO}_3^- = 14 \text{ mg/L}$, $\Delta = -8.29$) TDS contributes a

proportionally larger share of the index, yielding the highest observed sensitivity, yet the classification remains unchanged (Good). W042-1003 ($\text{NO}_3^- = 84 \text{ mg/L}$, HR-GWQI = 101.08) represents the sole classification change observed upon TDS removal ($\Delta = -1.19$, Poor $\rightarrow$ Good), as this borderline well is located only 1.08 points above the Good/Poor threshold. Critically, its nitrate concentration exceeds the WHO guideline of 50 mg/L by a factor of 1.68, independently confirming its public health significance regardless of its composite index classification. This pattern confirms that the AHP-derived weighting scheme functions as intended: it structurally amplifies the contribution of health-critical parameters particularly NO_3^- while attenuating the relative influence of mineralization indicators such as TDS, ensuring that HR-GWQI classifications are robust precisely in the wells where public health risk is highest. Compared with the Horton method, the AHP-derived weighting system substantially redefines parameter importance. Nitrate received the highest weight (32.00%), more than 2.7 times its relative weight in the conventional GWQI, reflecting its dominant anthropogenic origin and high toxicological significance. In contrast, Ca^{2+} , Mg^{2+} , HCO_3^- , and K^+ collectively accounted for only 12.11% of the total weight, consistent with their generally benign physiological roles (Minh et al., 2019; Badihi et al., 2023).

The HR-GWQI produced values ranging from 20.94 to 144.73 (mean: 83.42 ± 29.18). Wells W021-417, W042-439, and W042-1003 were reclassified from Good to Poor, while W021-893 was correctly upgraded from Poor to Good. Consequently, the proportion of wells suitable for drinking decreased from 78.3% under the GWQI to 69.6% under the HR-GWQI.

Spatial analysis based on the distribution of monitoring points further showed that the expansion of “Poor” groundwater quality zones under HR-GWQI is concentrated in the northeastern coastal fringe and western peripheral sectors of the plain, areas strongly affected by agricultural nitrogen loading, saline intrusion, and intensive groundwater extraction, as also identified by Ferdjouni et al. (2025), Aziez et al. (2025), and Khelfi et al. (2025). These findings are consistent with previous AHP-integrated WQI studies conducted in Pakistan and India, where health-risk-oriented weighting schemes produced more conservative and realistic groundwater quality classifications (Naz et al., 2024; Bisen and Singh, 2026). Nevertheless, several limitations remain, including the single dry-season sampling campaign, the exclusion of trace metals and microbiological indicators, reliance on a single-expert AHP matrix, and the absence of uncertainty analysis. Future studies should therefore incorporate multi-season monitoring, expanded parameter datasets, multi-expert validation approaches, and uncertainty propagation methods to improve the robustness and transferability of the HR-GWQI framework for semi-arid nitrate-contaminated aquifer systems across North Africa and the Mediterranean Basin.

Conclusion

This paper demonstrates that transitioning from a conventional GWQI to an AHP-calibrated HR-GWQI represents a meaningful advancement in aquifer health-risk assessment. Applied to the Mitidja Plain alluvial aquifer, the conventional index classified 78.3% of wells as suitable for drinking, yet systematically underestimated nitrate contamination by arithmetically diluting its contribution alongside toxicologically disparate parameters. The HR-GWQI corrects this structural bias, reducing the proportion of suitable wells to 69.6% and reclassifying wells with nitrate concentrations of 87–112 mg/L accordingly. Critically, the framework does not uniformly inflate index values; it also upgrades wells penalized by geochemically benign ions, confirming its selective, health-calibrated logic. GIS-based spatial distribution mapping of the monitoring points further reveals that the most significant reclassifications concentrate in the northeastern coastal fringe and western peripheral sector, where agricultural nitrogen loading, saline intrusion, and intensive abstraction converge. Three conclusions follow: the Horton GWQI remains valid for broad physicochemical screening but is insufficient as a standalone public health tool; AHP reweighting offers a rigorous and transferable recalibration mechanism applicable to analogous aquifer systems across North Africa and the Mediterranean; and GIS integration is essential for converting index results into spatially actionable governance information. Future work should incorporate multi-season campaigns, trace metals, microbiological indicators, Delphi-based expert validation, and Monte Carlo uncertainty analysis to further consolidate the scientific robustness of this approach.

A sensitivity analysis further demonstrated that the HR-GWQI is structurally most robust in the wells with the highest nitrate burden an emergent property of the AHP health-risk-oriented weighting scheme that reinforces its validity as a tool for groundwater governance in nitrate-contaminated aquifers.

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Author contributions

ZITOUNI Zahida B. led the conceptualization of the study, developed the research methodology, and supervised the overall project. HARTANI Tarik, serving as the supervisor, critically reviewed and proofread the manuscript. Both ZITOUNI Zahida B. and HARTANI Tarik were actively involved in data collection, data cleaning, analysis, and interpretation of the results. BOURAS-CHEKIREF Fatma-Zohra and AIDAOU Abdellah provided valuable input through their thorough review and approval of the final manuscript. All authors contributed to the manuscript and approved the final version for publication.

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AI use declaration statement

The authors used LLMs (such as Genspark and Claude) during the writing process to improve readability and grammar in some parts of the paper. All AI-generated suggestions and outputs, including references, were reviewed and verified by the authors for factual accuracy and correctness including references.

Competing interest

All authors declare no competing interests.

Additional information

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